

Bounded Differences Inequality

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1 Bounded Differences Inequality

We say that a function f satisfies the bounded difference property with parameters $(L_1, \dots, L_n)$ if, for each $k = 1, \dots, n$,

$$\sup_{x_1, \dots, x_n, x'_k} |f(x_1, \dots, x_k, \dots, x_n) - f(x_1, \dots, x_{k-1}, x'_k, x_{k+1}, \dots, x_n)| \leq L_k. \quad (1)$$

Corollary 1.1 (Bounded Differences Inequality). *Suppose that f satisfies the bounded difference property with $(L_1, \dots, L_n)$, and the random vector $X = (X_1, \dots, X_n)$ has independent components. Then for every $t \geq 0$,*

$$\mathbb{P}(|f(X) - \mathbb{E}f(X)| \geq t) \leq 2 \exp \left(-\frac{2t^2}{\sum_{k=1}^n L_k^2} \right). \quad (2)$$

Proof.

$$\begin{aligned} D_k &= \mathbb{E}[f(X) \mid X_1, \dots, X_k] - \mathbb{E}[f(X) \mid X_1, \dots, X_{k-1}] \\ &= \mathbb{E}_{X_{k+1}, \dots, X_n}[f(X)] - \mathbb{E}_{X_k, X_{k+1}, \dots, X_n}[f(X)] \\ &:= \mathbb{E}_{k+1}[f(X)] - \mathbb{E}_k[f(X)]. \end{aligned}$$

Then $D_k \in \mathcal{F}_k$, and

$$\begin{aligned} \mathbb{E}[D_k \mid \mathcal{F}_{k-1}] &= \mathbb{E}[\mathbb{E}[f(X) \mid \mathcal{F}_k] - \mathbb{E}[f(X) \mid \mathcal{F}_{k-1}] \mid \mathcal{F}_{k-1}] \\ &= \mathbb{E}[f(X) \mid \mathcal{F}_{k-1}] - \mathbb{E}[f(X) \mid \mathcal{F}_{k-1}] \\ &= 0 \quad \text{a.s.} \end{aligned}$$

Thus, $(D_k)_{k=1}^n$ is a martingale difference sequence. Now,

$$\begin{aligned} D_n &= \mathbb{E}_{n+1}f(X) - \mathbb{E}_n[f(X)], \\ D_1 &= \mathbb{E}_2f(X) - \mathbb{E}_1f(X) = \mathbb{E}_2f(X) - \mathbb{E}f(X), \\ \text{and so, } f(X) - \mathbb{E}f(X) &= \sum_{k=1}^n D_k. \end{aligned}$$

Next,

$$\begin{aligned} D_k &= \mathbb{E}_{k+1}[f(X_1, \dots, X_{k-1}, X_k, X_{k+1}, \dots, X_n)] - \mathbb{E}_k[f(X_1, \dots, X_{k-1}, X_k, X_{k+1}, \dots, X_n)] \\ &\leq \sup_x \mathbb{E}_{k+1}[f(X_1, \dots, X_{k-1}, x, X_{k+1}, \dots, X_n)] - \mathbb{E}_k[f(X_1, \dots, X_{k-1}, X_k, X_{k+1}, \dots, X_n)] \\ &:= B_k, \end{aligned}$$

where $B_k \in \mathcal{F}_{k-1}$. Similarly,

$$\begin{aligned} D_k &\geq \inf_x \mathbb{E}_{k+1}[f(X_1, \dots, X_{k-1}, x, X_{k+1}, \dots, X_n)] - \mathbb{E}_k[f(X_1, \dots, X_{k-1}, X_k, X_{k+1}, \dots, X_n)] \\ &:= A_k, \end{aligned}$$

where $A_k \in \mathcal{F}_{k-1}$. Then, we have

$$\begin{aligned} B_k - A_k &= \sup_x \mathbb{E}_{k+1} [f(X_1, \dots, X_{k-1}, x, X_{k+1}, \dots, X_n)] - \inf_{x'} \mathbb{E}_{k+1} [f(X_1, \dots, X_{k-1}, x', X_{k+1}, \dots, X_n)] \\ &\leq \sup_{x_1, \dots, x_{k-1}, x_k, x'_k, x_{k+1}, \dots, x_n} |f(x_1, \dots, x_k, \dots, x_n) - f(x_1, \dots, x_{k-1}, x'_k, x_{k+1}, \dots, x_n)| \\ &\leq L_k, \end{aligned}$$

using (1). Therefore, by Azuma–Hoeffding,

$$\mathbb{P} \left(\left| \sum_{k=1}^n D_k \right| \geq t \right) = \mathbb{P}(|f(X) - \mathbb{E}f(X)| \geq t) \leq 2 \exp \left(-\frac{2t^2}{\sum_{k=1}^n L_k^2} \right).$$

□

Corollary 1.2 (Azuma–Hoeffding). *Let $\{(D_k, \mathcal{F}_k)\}_{k=1}^\infty$ be a martingale difference sequence and suppose $\mathbb{E}[D_k | \mathcal{F}_{k-1}] \in [a_k, b_k]$, where $a_k, b_k \in \mathcal{F}_{k-1}$ for all $k = 1, \dots, n$. Assume that $b_k - a_k \leq L_k$ is a constant. Then, for all $t \geq 0$,*

$$\mathbb{P} \left(\left| \sum_{k=1}^n D_k \right| \geq t \right) \leq 2 \exp \left(-\frac{2t^2}{\sum_{k=1}^n L_k^2} \right). \quad (3)$$

2 Examples

Example 2.1 (U-statistics). Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a symmetric function of its arguments. Given i.i.d. sequence X_k , $k \geq 1$ of random variables, the quantity

$$U := \frac{1}{\binom{n}{2}} \sum_{1 \leq j < k \leq n} g(X_j, X_k) \quad (4)$$

is known as a pairwise U-statistic.

For example, suppose we want to estimate the mean absolute pairwise deviation $\mathbb{E}[|X_1 - X_2|]$. Then U with $g(x, y) = |x - y|$ is one estimate for it.

Since the X_i are identically distributed,

$$\mathbb{E}U = \frac{1}{\binom{n}{2}} \sum_{1 \leq j < k \leq n} \mathbb{E}g(X_j, X_k) = \frac{1}{\binom{n}{2}} \sum_{1 \leq j < k \leq n} \mathbb{E}g(X_1, X_2) = \mathbb{E}g(X_1, X_2).$$

Suppose $\|g\|_\infty \leq b$. Consider $U = f(X_1, \dots, X_n)$. Then

$$\begin{aligned} &|f(X_1, \dots, X_n) - f(X_1, \dots, X_{k-1}, X'_k, X_{k+1}, \dots, X_n)| \\ &= \left| \frac{1}{\binom{n}{2}} \sum_{1 \leq j < s \leq n} g(X_j, X_s) - \frac{1}{\binom{n}{2}} \sum_{1 \leq j < s \leq n} g(X'_j, X'_s) \right| \quad [\text{where } X'_j = X_j, j = 1, \dots, k-1, k+1, \dots, n], \\ &= \frac{1}{\binom{n}{2}} \left| \sum_{1 \leq j < s \leq n} g(X_j, X_s) - \sum_{1 \leq j < s \leq n} g(X'_j, X'_s) \right| \\ &\leq \frac{1}{\binom{n}{2}} \sum_{1 \leq j < s \leq n} |g(X_j, X_s) - g(X'_j, X'_s)| \\ &\leq \frac{1}{\binom{n}{2}} |2b(n - k + k - 1)| \\ &= \frac{4b}{n}. \end{aligned}$$

So the bounded differences property holds with parameter $L_k = \frac{4b}{n}$ in each coordinate. Therefore, the bounded difference inequality gives

$$\mathbb{P}(|U - \mathbb{E}U| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{k=1}^n (4b/n)^2}\right) = 2 \exp\left(-\frac{nt^2}{8b^2}\right). \quad (5)$$

Example 2.2 (Rademacher Complexity). Let $\{\varepsilon_k\}_{k=1}^n$ be an i.i.d. sequence of Rademacher variables. Given a collection of vectors $\mathcal{A} \subset \mathbb{R}^n$, define the random variable

$$Z(\mathcal{A}) = Z := \sup_{a \in \mathcal{A}} \langle a, \varepsilon \rangle = \sup_{a \in \mathcal{A}} \sum_{i=1}^n a_i \varepsilon_i. \quad (6)$$

Then $R(\mathcal{A}) := \mathbb{E}Z(\mathcal{A})$ is the Rademacher complexity of $\mathcal{A}$.

Now let us establish $Z(\mathcal{A}) = f(\varepsilon_1, \dots, \varepsilon_n)$ is sub-Gaussian. Given two Rademacher vectors $\varepsilon, \varepsilon' \in \{-1, +1\}^n$, and $\varepsilon'_i = \varepsilon_i$ for $i \neq k$, we have

$$\begin{aligned} |f(\varepsilon_1, \dots, \varepsilon_n) - f(\varepsilon'_1, \dots, \varepsilon'_n)| &= \left| \sup_{a \in \mathcal{A}} \langle a, \varepsilon \rangle - \sup_{a \in \mathcal{A}} \langle a, \varepsilon' \rangle \right| \\ &\leq \sup_{a \in \mathcal{A}} |\langle a, \varepsilon \rangle - \langle a, \varepsilon' \rangle| \quad (\text{by Lemma 2.3}) \\ &= \sup_{a \in \mathcal{A}} |a_k \varepsilon_k - a_k \varepsilon'_k| \\ &= \sup_{a \in \mathcal{A}} |a_k| |\varepsilon_k - \varepsilon'_k| \\ &\leq 2 \sup_{a \in \mathcal{A}} |a_k|. \end{aligned}$$

Consequently, the bounded difference inequality yields

$$\mathbb{P}(|Z(\mathcal{A}) - R(\mathcal{A})| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{k=1}^n 4 \sup_{a \in \mathcal{A}} a_k^2}\right) = 2 \exp\left(-\frac{t^2}{2 \sum_{k=1}^n \sup_{a \in \mathcal{A}} a_k^2}\right).$$

Thus, $Z(\mathcal{A})$ is sub-Gaussian with parameter $\sqrt{2 \sum_{k=1}^n \sup_{a \in \mathcal{A}} a_k^2}$.

Lemma 2.3. *The absolute difference of the suprema of two functions is bounded by the supremum of their pointwise absolute difference*

$$\left| \sup_{x \in B} f(x) - \sup_{x \in B} g(x) \right| \leq \sup_{x \in B} |f(x) - g(x)|.$$

References

- [1] M. J. Wainwright, *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*, Cambridge University Press, 2019.
- [2] R. Vershynin, *High-Dimensional Probability: An Introduction with Applications in Data Science*, Cambridge University Press, 2018.